

# AFAE - Numerical validation using the CADNA library

Delmotte Aurélien

Zhang Xiru

October 2024

## Contents

|          |                                                                  |          |
|----------|------------------------------------------------------------------|----------|
| <b>1</b> | <b>Exercise 1 Rump's Example</b>                                 | <b>2</b> |
| 1.1      | Question 1: Single Precision Calculation . . . . .               | 2        |
| 1.2      | Question 2: Double Precision Calculation . . . . .               | 2        |
| 1.3      | Question 3: Implementation with CADNA . . . . .                  | 2        |
| <b>2</b> | <b>Exercise 2 Hilbert Matrix Determinant Calculation</b>         | <b>2</b> |
| 2.1      | Question 1: Gaussian Elimination without CADNA . . . . .         | 2        |
| 2.2      | Question 2: Gaussian Elimination with CADNA . . . . .            | 2        |
| <b>3</b> | <b>Exercise 3 Muller's Example</b>                               | <b>3</b> |
| 3.1      | Question 1: Iterations without CADNA . . . . .                   | 3        |
| 3.2      | Question 2: Iterations with CADNA . . . . .                      | 3        |
| <b>4</b> | <b>Exercise 4 Newton Iteration</b>                               | <b>3</b> |
| 4.1      | Question 1: Newton Iteration Without CADNA . . . . .             | 3        |
| 4.2      | Question 2: Newton Iteration With CADNA . . . . .                | 3        |
| <b>5</b> | <b>Exercise 5 Gaussian Elimination</b>                           | <b>4</b> |
| 5.1      | Question 1: Gaussian Elimination Without CADNA . . . . .         | 4        |
| 5.2      | Question 2: Gaussian Elimination With CADNA . . . . .            | 4        |
| <b>6</b> | <b>Exercise 6 Jacobi Iteration</b>                               | <b>4</b> |
| 6.1      | Question 1: Question 1: Jacobi Iteration Without CADNA . . . . . | 4        |
| 6.2      | Question 2: Jacobi Iteration With CADNA . . . . .                | 4        |
| <b>7</b> | <b>Exercise 7 Logistic Iteration</b>                             | <b>4</b> |
| 7.1      | Question 1: Without CADNA . . . . .                              | 4        |
| 7.2      | Question 2: With CADNA . . . . .                                 | 5        |

## 1 Exercise 1 Rump's Example

### 1.1 Question 1: Single Precision Calculation

The computed result using single precision was:

$$\text{res} = 7.08931176699221 \times 10^{29}$$

This result significantly deviates from the exact value. This discrepancy highlights the limitations of single precision floats in handling calculations with large numerical ranges and sensitive operations.

### 1.2 Question 2: Double Precision Calculation

After modifying the variable types from float to double, the computed result was:

$$\text{res} = 5.06997231650861 \times 10^{20}$$

The result demonstrates that changing from single to double precision improved the accuracy but did not come close to the exact value, highlighting the challenges and limitations of floating-point arithmetic in numerical computing.

### 1.3 Question 3: Implementation with CADNA

Both programs return the result marked as "@.0", indicating that the result retains no accurate data. The accompanying message from the CADNA reports:

There is 1 numerical instability and 1 loss(es) of accuracy due to cancellation(s).

This output confirms that the computational answers are wholly inaccurate, as evidenced by the significant discrepancies when compared with the exact results.

## 2 Exercise 2 Hilbert Matrix Determinant Calculation

### 2.1 Question 1: Gaussian Elimination without CADNA

```
Pivot 0 = +1.000000000000000e+00
Pivot 1 = +8.333333333333331e-02
...
Pivot 10 = +1.399228141909059e-12
Determinant: +3.028594215199785e-65
```

This result, while very small, aligns closely with the theoretical value of the determinant, suggesting that the calculation was handled with sufficient numerical accuracy, which seems to be accurate up to the first 2 significant decimal digits.

### 2.2 Question 2: Gaussian Elimination with CADNA

```
Pivot 0 = 0.100000000000000e+001
Pivot 1 = 0.833333333333333e-001
...
Pivot 10 = 0.13e-011
Determinant: 0.30e-64
```

The determinant is accurate up to the first two significant decimal digits.

## 3 Exercise 3 Muller's Example

### 3.1 Question 1: Iterations without CADNA

Despite the computations stabilizing at a value of 100 for  $U(29)$  and  $U(30)$ :

$$\begin{aligned}U(29) &= +1.000000000000000e + 02, \\U(30) &= +1.000000000000000e + 02,\end{aligned}$$

these results are incorrect given that the theoretical limit of the sequence should be 6. This discrepancy indicates significant numerical errors or algorithmic instability not accounted for in the basic implementation of the sequence.

### 3.2 Question 2: Iterations with CADNA

While the implementation with CADNA also did not yield the correct sequence limit, it provided valuable insights into the nature of the computational errors:

- CADNA reported 8 numerical instabilities, including 6 unstable divisions and 2 unstable multiplications. This highlights the sequence's susceptibility to errors due to critical operations that are numerically unstable.
- Detailed debugging with additional print statements revealed that accurate data was lost around  $U(13)$ , traceable to a cancellation error when reducing the expression  $111 - 10X = 10 - 0X \rightarrow XX$ .
- Interestingly, the sequence showed some recovery of accuracy in the final iterations. This recovery could be attributed to the constant term 111 in the sequence formula, which might have partially absorbed the impact of other terms, leading to the stabilization at 100. This stabilization is similar to what would be observed if the sequence were initialized with  $b = 111$ , suggesting that the constant term dominates the sequence behavior under certain numerical conditions.

## 4 Exercise 4 Newton Iteration

### 4.1 Question 1: Newton Iteration Without CADNA

The sequence converges towards  $4.28571431489566e-01$ , after 26 iterations meeting the stopping criterion of  $|x_n - x_{n-1}| < 10^{-12}$ . This result suggests that, within the constraints of the specified precision and algorithmic approach (Newton's method), the calculation successfully converges to a consistent value.

### 4.2 Question 2: Newton Iteration With CADNA

The initial implementation with CADNA revealed a significant number of numerical instabilities:

- The sequence initially converged similarly to the non-CADNA version but eventually resulted in numerical instabilities, with the CADNA report showing:
  - 476 numerical instabilities, including unstable divisions, branching, intrinsic functions, and cancellations.
- Despite these early signs of instability, the computation technically continued until the 100th iteration, the results from the 23rd iteration onwards are marked as "@.0", indicating significant numerical instabilities that question the reliability of subsequent computations.

Changing the stopping criterion to exact equality ( $x == y$ ) showed that the method could only proceed up to the 24th iteration before encountering pervasive numerical instabilities:

- The iterations did not significantly extend beyond what was observed under the original criterion. The series stabilized around 0.4285714 but with continuous numerical instability annotations.
- The CADNA analysis under this condition reported 48 losses of accuracy due to cancellations, demonstrating how a more stringent stopping criterion quickly highlights the limitations of floating-point arithmetic.

## 5 Exercise 5 Gaussian Elimination

### 5.1 Question 1: Gaussian Elimination Without CADNA

The results obtained from the Gaussian elimination method are presented as follows:

Computed solutions:  $(6.261988 \times 10^{01}, -8.953979, 0.000000, 1.000000)$

Exact solutions:  $(1, 1, 10^{-8}, 1)$

These results indicate substantial discrepancies between the computed values and the exact solutions, particularly for the first two variables.

### 5.2 Question 2: Gaussian Elimination With CADNA

The results obtained from the Gaussian elimination method using CADNA are presented as follows:

Computed solutions with CADNA:  $(0.99, 1.00, 9.99999 \times 10^{-8}, 1.00)$

Exact solutions:  $(1, 1, 10^{-8}, 1)$

These results show a significant improvement in accuracy compared to the non-CADNA method, with the computed values closely matching the exact solutions.

## 6 Exercise 6 Jacobi Iteration

### 6.1 Question 1: Question 1: Jacobi Iteration Without CADNA

Running the 'jacobi.c' program with the stopping criterion  $\epsilon = 10^{-4}$  resulted in 1000 iterations, suggesting a high degree of refinement to achieve the desired precision. However, adjusting the epsilon to  $10^{-2}$  significantly reduced the number of iterations to 28, showing that a less stringent stopping criterion can lead to fewer iterations but at the cost of accuracy. The results indicate that:

For  $\epsilon = 10^{-4}$ , errors ranged from  $-0.006396178$  to  $+0.005910804$ .

For  $\epsilon = 10^{-2}$ , errors were notably larger, up to  $-0.02645079$ .

### 6.2 Question 2: Jacobi Iteration With CADNA

After integrating CADNA into the Jacobi iteration, the algorithm completed in 23 iterations, significantly fewer than the 1000 iterations in the non-CADNA version. This reduction is due to the early detection of convergence through the monitoring of the anorm value.

In the final iteration (23rd iteration), the value of 'anorm' was found to be:

`anorm = 0.92E-001`

This value indicates that the largest change between the solutions in the 23rd and 22nd iterations is approximately 0.092. Since this value is smaller than the threshold  $\epsilon = 1 \times 10^{-1}$ , the algorithm stopped at this point, considering the solution sufficiently converged.

## 7 Exercise 7 Logistic Iteration

### 7.1 Question 1: Without CADNA

The results obtained from the logistic iteration using two mathematically equivalent expressions show slight differences in their computed values. The iteration results for both expressions diverge slightly as the number of iterations increases, particularly by iteration 200, highlighting the accumulation of numerical errors in practice. The first expression is  $x = a \cdot 0.25 - a \cdot (x - 0.5)^2$ :

```

i= 50  x= 3.875950636897043e-01
i=100  x= 4.564083376315182e-01
i=150  x= 4.261690127424104e-01
i=200  x= 3.762927173332673e-01
last iterate:
i=200  x= 3.762927173332673e-01

```

The second expression is  $x = ax(1 - x)$ :

```

i= 50  x= 3.875950637036066e-01
i=100  x= 4.564074671150947e-01
i=150  x= 4.062664991348743e-01
i=200  x= 5.761024866343428e-01
last iterate:
i=200  x= 5.761024866343428e-01

```

These results indicate that despite both expressions being mathematically equivalent, numerical errors accumulate, leading to different results after 200 iterations.

## 7.2 Question 2: With CADNA

Integrating CADNA into the logistic iteration revealed 54 numerical instabilities, specifically 16 unstable multiplications and 38 unstable power functions. In this iteration, two different expressions were used to calculate  $x$  and  $y$ :

- $x = a \cdot x \cdot (1 - x)$
- $y = a \cdot 0.25 - a \cdot (x - 0.5)^2$

By iteration 157, the results became unreliable, with CADNA marking one of the outputs as "@.0", signaling the loss of precision. Below are the results from running the logistic iteration:

```

i= 50  x= 0.3875950636E+000  y= 0.8545144690E+000
i=100  x= 0.45640E+000      y= 0.893159E+000
i=150  x= 0.4E+000         y= 0.8E+000
i=200  x= @.0              y= 0.8E+000
last iterate:
i=200  x= @.0              y= 0.8E+000

```

The CADNA software detected 54 numerical instabilities:

```

16 UNSTABLE MULTIPLICATION(S)
38 UNSTABLE POWER FUNCTION(S)

```

**Stopping Criterion to Avoid Useless Iterations:** To avoid unnecessary iterations, the stopping criterion should be set based on detecting significant numerical instability or when the difference between two consecutive iterations becomes smaller than a predefined threshold  $\epsilon$ , such as  $10^{-8}$ . Additionally, CADNA's detection of instabilities can be leveraged to halt the iterations when precision degrades beyond an acceptable level.